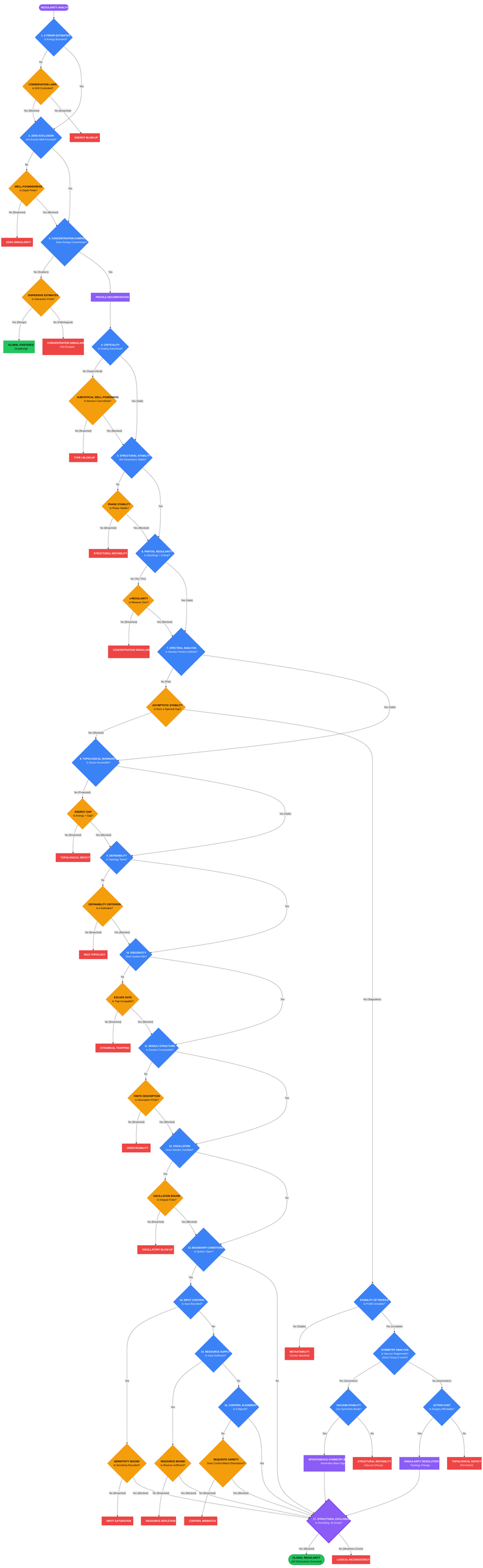
REGULARITY ANALYSIS
1. A PRIORI ESTIMATES
Is Energy Bounded?
No
Yes
CONSERVATION LAWS
Is Drift Controlled?
Yes (Blocked)
No (Breached)
ENERGY BLOW-UP
2. ZENO EXCLUSION
Are Events Well-Founded?
No
Yes
WELL-FOUNDEDNESS
Is Depth Finite?
No (Breached)
Yes (Blocked)
ZENO SINGULARITY
3. CONCENTRATION COMP.
Does Energy Concentrate?
No (Scatters)
Yes
DISPERSIVE ESTIMATES
Is Interaction Finite?
Yes (Benign)
No (Pathological)
GLOBAL EXISTENCE
(Scattering)
CONCENTRATION SINGULARITY
(Old Escape)
PROFILE DECOMPOSITION
4. CRITICALITY
Is Scaling Subcritical?
No (Supercritical)
Yes (Safe)
SUBCRITICAL WELL-POSEDNESS
Is Renorm Cost Infinite?
No (Breached)
Yes (Blocked)
TYPE I BLOW-UP
5. STRUCTURAL STABILITY
Are Parameters Stable?
No
Yes
PHASE STABILITY
Is Phase Stable?
No (Breached)
Yes (Blocked)
STRUCTURAL INSTABILITY
6. PARTIAL REGULARITY
Is dimSing < Critical?
No (Too Thin)
ε-REGULARITY
Is Measure Zero?
Yes (Safe)
No (Breached)
Yes (Blocked)
CONCENTRATION SINGULARITY
7. SPECTRAL ANALYSIS
Is Hessian Positive Definite?
No (Flat)
Yes (Safe)
ASYMPTOTIC STABILITY
Is there a Spectral Gap?
Yes (Blocked)
No (Diagnostics)
8. TOPOLOGICAL INVARIANTS
Is Sector Accessible?
No (Protected)
Yes (Safe)
ENERGY GAP
Is Energy > Gap?
No (Breached)
Yes (Blocked)
TOPOLOGICAL DEFECT
9. DEFINABILITY
Is Topology Tame?
No
Yes
DEFINABILITY CRITERION
Is it Definable?
No (Breached)
Yes (Blocked)
WILD TOPOLOGY
10. ERGODICITY
Does System Mix?
No
Yes
ESCAPE RATE
Is Trap Escapable?
No (Breached)
Yes (Blocked)
DYNAMICAL TRAPPING
11. MODULI STRUCTURE
Is Solution Computable?
No
Yes
FINITE DESCRIPTION
Is Description Finite?
No (Breached)
Yes (Blocked)
UNDECIDABILITY
12. OSCILLATION
Does Solution Oscillate?
Yes
No
OSCILLATION BOUND
Is Integral Finite?
No (Breached)
Yes (Blocked)
OSCILLATORY BLOW-UP
13. BOUNDARY CONDITIONS
Is System Open?
Yes
No
14. INPUT CONTROL
Is Input Accessible?
Yes
No
15. RESOURCE SUPPLY
Is Input Sufficient?
Yes
No
16. CONTROL ALIGNMENT
Is it Aligned?
Yes
No
SENSITIVITY BOUND
Is Sensitivity Bounded?
No (Breached)
Yes (Blocked)
INPUT SATURATION
RESOURCE BOUND
Is Reserve Sufficient?
No (Breached)
Yes (Blocked)
RESOURCE DEPLETION
REQUISITE VARIETY
Does Control Match Disturbance?
No (Breached)
Yes (Blocked)
CONTROL MISMATCH
STABILITY OF PROFILE
Is Profile Unstable?
No (Stable)
Yes (Unstable)
METASTABILITY
(Centre Manifold)
SYMMETRY ANALYSIS
Is Vacuum Degenerate?
(Does Group G exist?)
Yes (Symmetric)
No (Asymmetric)
VACUUM STABILITY
Can Symmetry Break?
Yes
No
SPONTANEOUS SYMMETRY BREAKING
Generates Mass Gap
STRUCTURAL INSTABILITY
(Vacuum Decay)
ACTION COST
Is Surgery Affordable?
Yes
No
SINGULARITY RESOLUTION
Topology Change
TOPOLOGICAL DEFECT
(Protected)
17. STRUCTURAL EXCLUSION
Is HomDim(S, G) Empty?
Yes (Blocked)
No (Morphism Exists)
GLOBAL REGULARITY
(All Obstructions Excluded)
LOGICAL INCONSISTENCY